



Lecture Outline

- Classification: Why not Linear Regression?
- Binary Response & Logistic Regression
- Estimating the Simple Logistic Model
- Classification using the Logistic Model
- Multiple Logistic Regression

Logistic Regression

Logistic Regression addresses the problem of estimating a probability, $P(y = 1)$, to be outside the range of $[0,1]$. The logistic regression model uses a function, called the *logistic* function, to model $P(y = 1)$:

$$P(Y = 1) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

Logistic Regression

As a result, the model will predict $P(y = 1)$ with an S -shaped curve, which is the general shape of the logistic function.

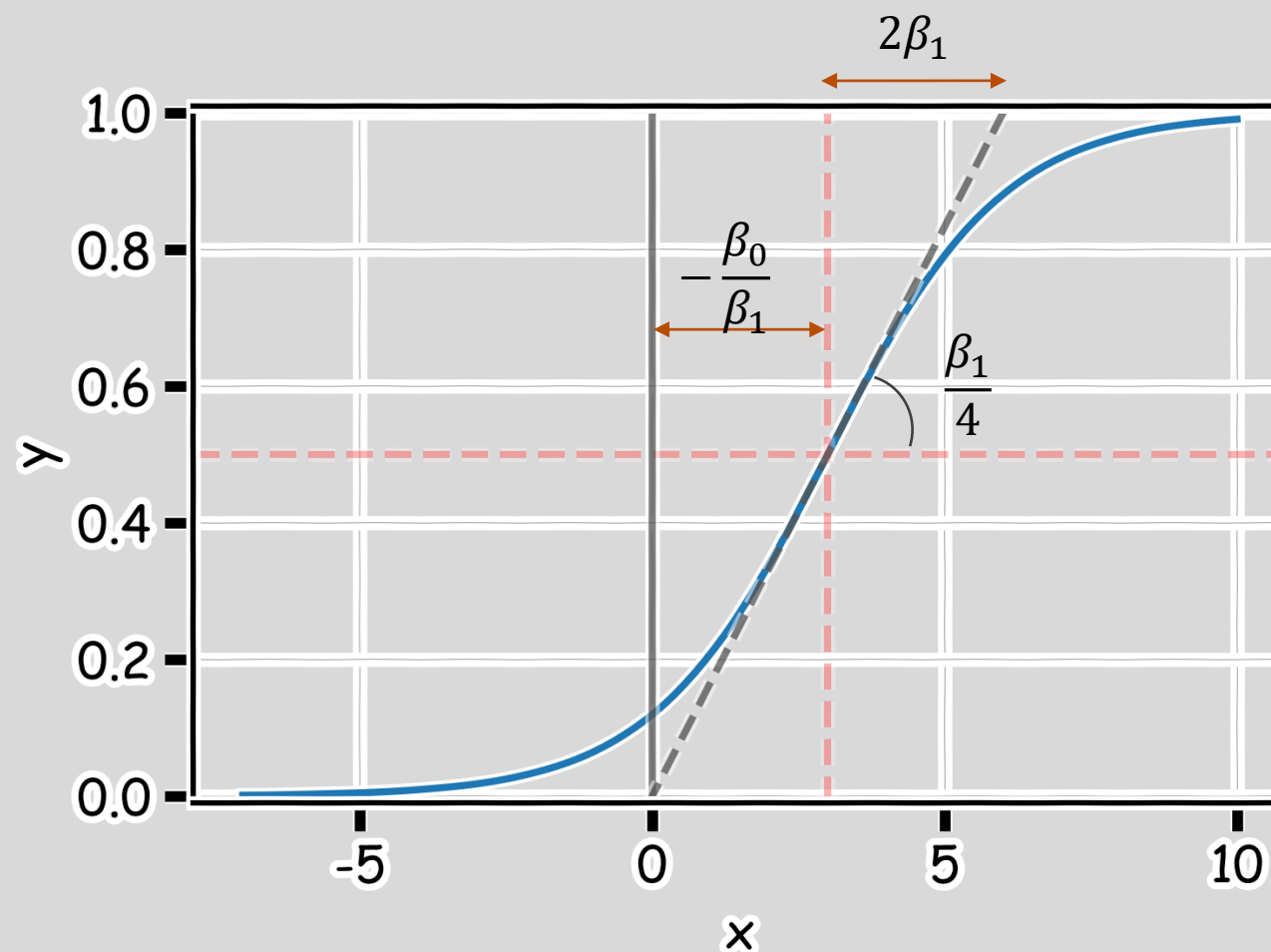
β_0 shifts the curve right or left by $c = -\frac{\beta_0}{\beta_1}$.

β_1 controls how steep the S -shaped curve is. Distance from $\frac{1}{2}$ to almost 1 or $\frac{1}{2}$ to almost 0 to $\frac{1}{2}$ is $\frac{2}{\beta_1}$

Note: if β_1 is positive, then the predicted $P(y = 1)$ goes from zero for small values of X to one for large values of X and if β_1 is negative, then the $P(y = 1)$ has opposite association.

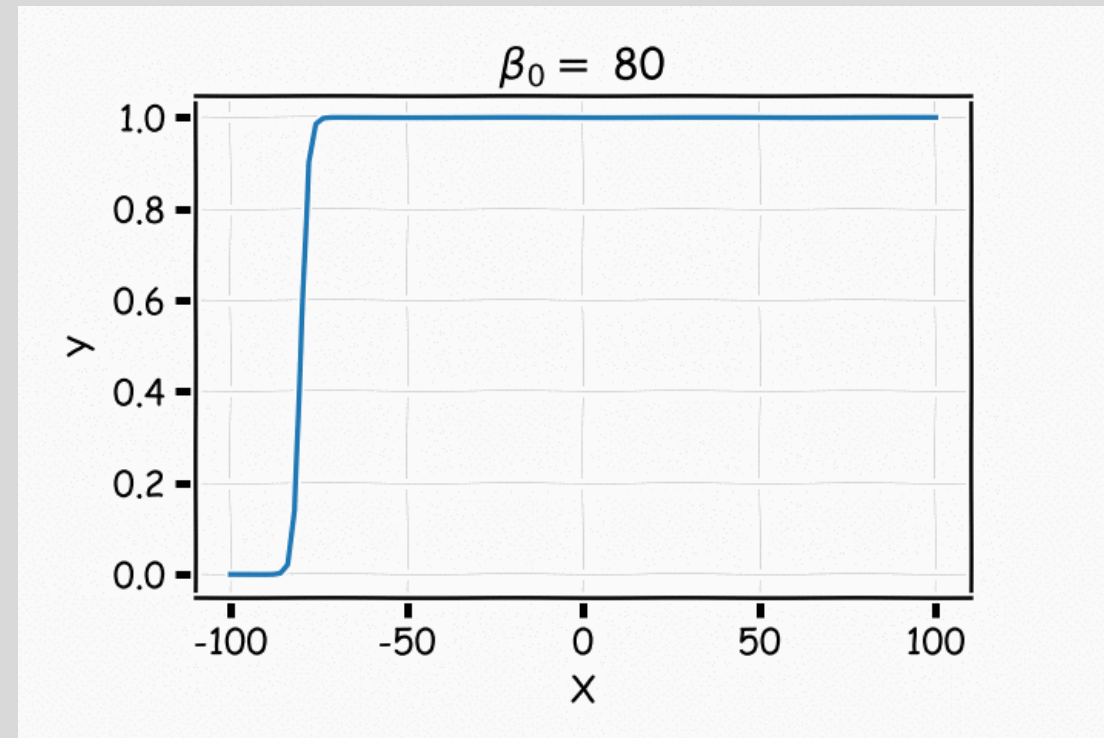


Logistic Regression



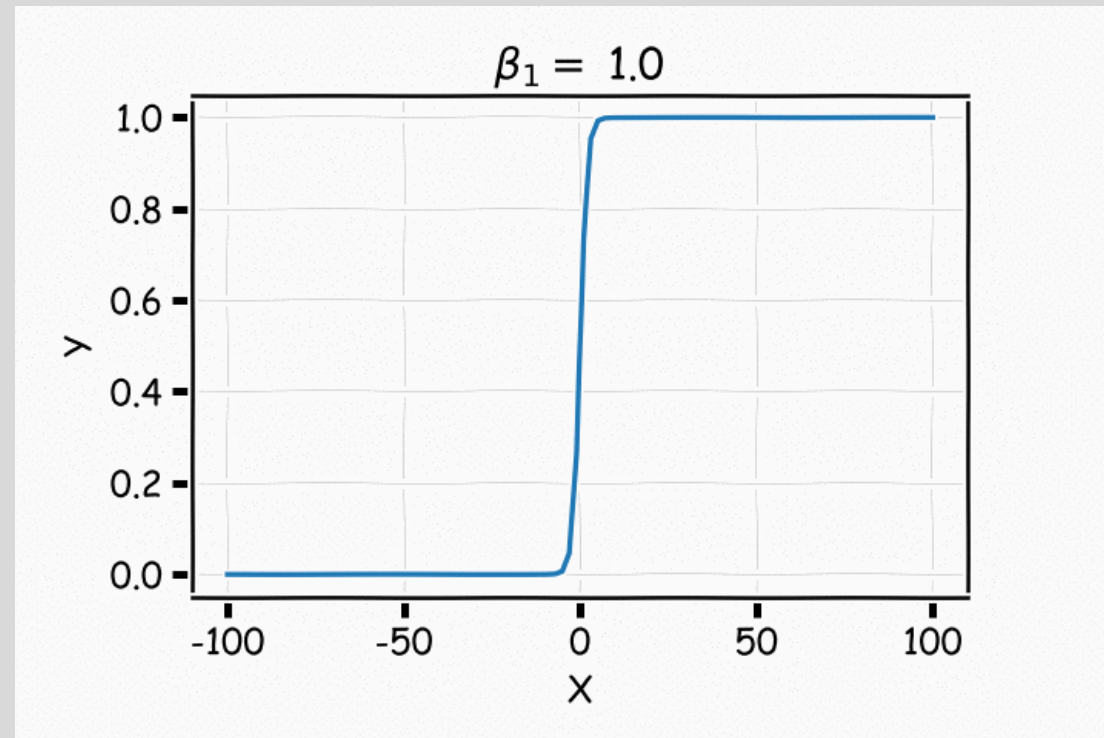
Logistic Regression

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$



Logistic Regression

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$



Logistic Regression

With a little bit of algebraic work, the logistic model can be rewritten as:

$$\ln \left(\frac{P(Y = 1)}{1 - P(Y = 1)} \right) = \beta_0 + \beta_1 X.$$

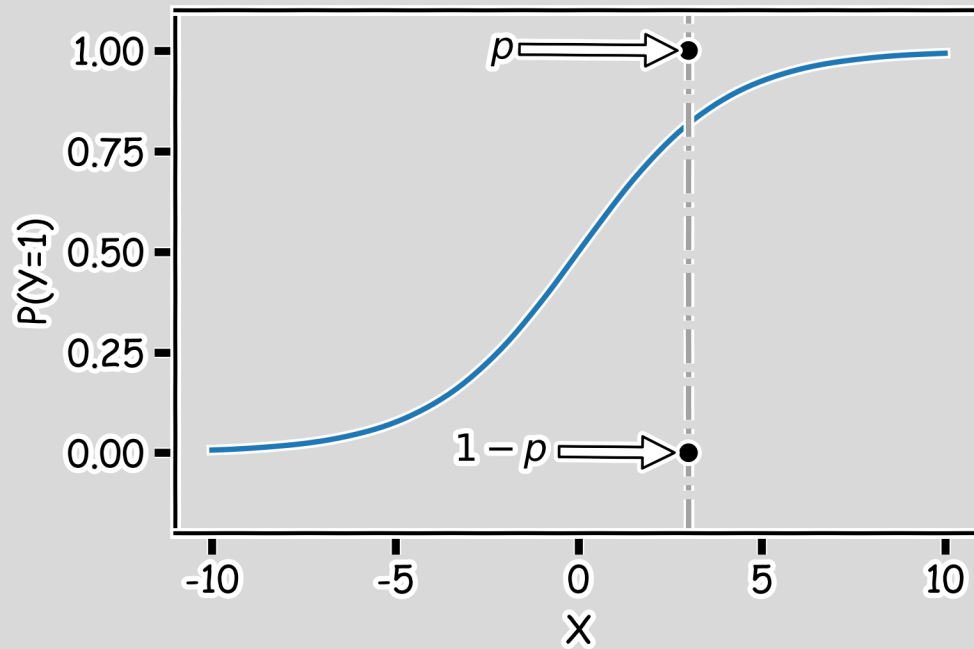
The value inside the natural log function $\frac{P(Y=1)}{1-P(Y=1)}$, is called the *odds*, thus logistic regression is said to model the *log-odds* with a linear function of the predictors or features, X . This gives us the natural interpretation of the estimates similar to linear regression: a one unit change in X is associated with a β_1 change in the log-odds of $Y = 1$; or better yet, a one unit change in X is associated with an e^{β_1} change in the odds that $Y = 1$.



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Estimation in Logistic Regression



Probability $Y = 1$: p

Probability $Y = 0$: $1 - p$

$$P(Y = y) = p^y (1 - p)^{(1-y)}$$

where:

$p = P(Y = 1|X = x)$ and therefore p depends on X .

Thus not every p is the same for each individual measurement.

Likelihood

The likelihood of a single observation for p given x and y is:

$$L(p_i|Y_i) = P(Y_i = y_i) = p_i^{y_i}(1 - p_i)^{1-y_i}$$

Given the observations are independent, what is the likelihood function for p ?

$$L(p|Y) = \prod_i P(Y_i = y_i) = \prod_i p_i^{y_i}(1 - p_i)^{1-y_i}$$

$$l(p|Y) = -\log L(p|Y) = -\sum_i y_i \log p_i + (1 - y_i) \log(1 - p_i)$$

Loss Function

$$l(p|Y) = - \sum_i \left[y_i \log \frac{1}{1 + e^{-\beta X_i}} + (1 - y_i) \log \left(1 - \frac{1}{1 + e^{-\beta X_i}} \right) \right]$$

How do we minimize this?

Differentiate, equate to zero and solve for it!

But this looks messy! It will not necessarily have a closed form solution.

So how do we determine the parameter estimates?

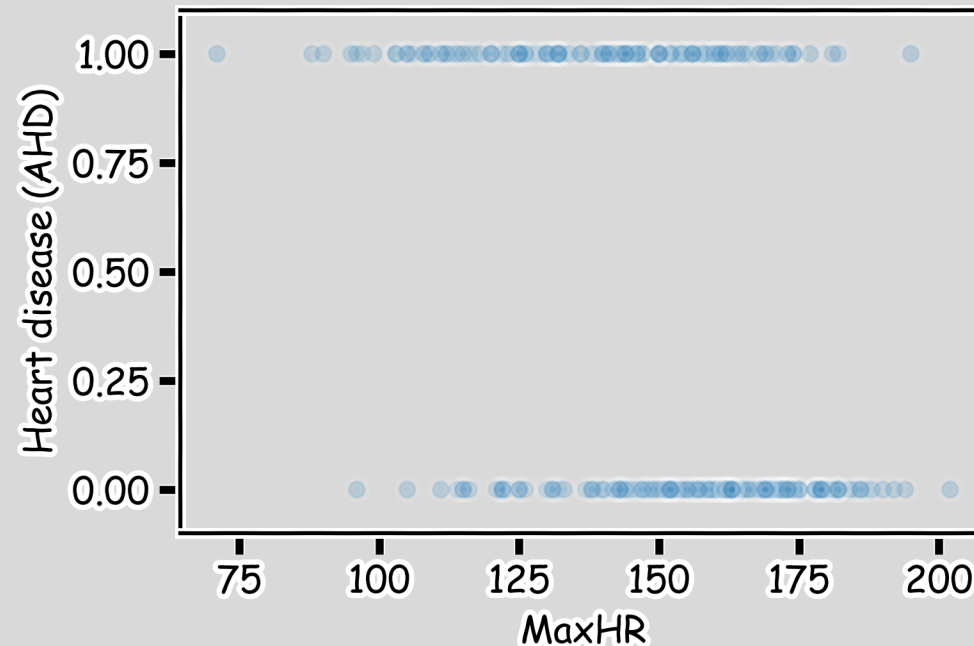
Through an iterative approach

(we will talk about this *at length* in future lectures).

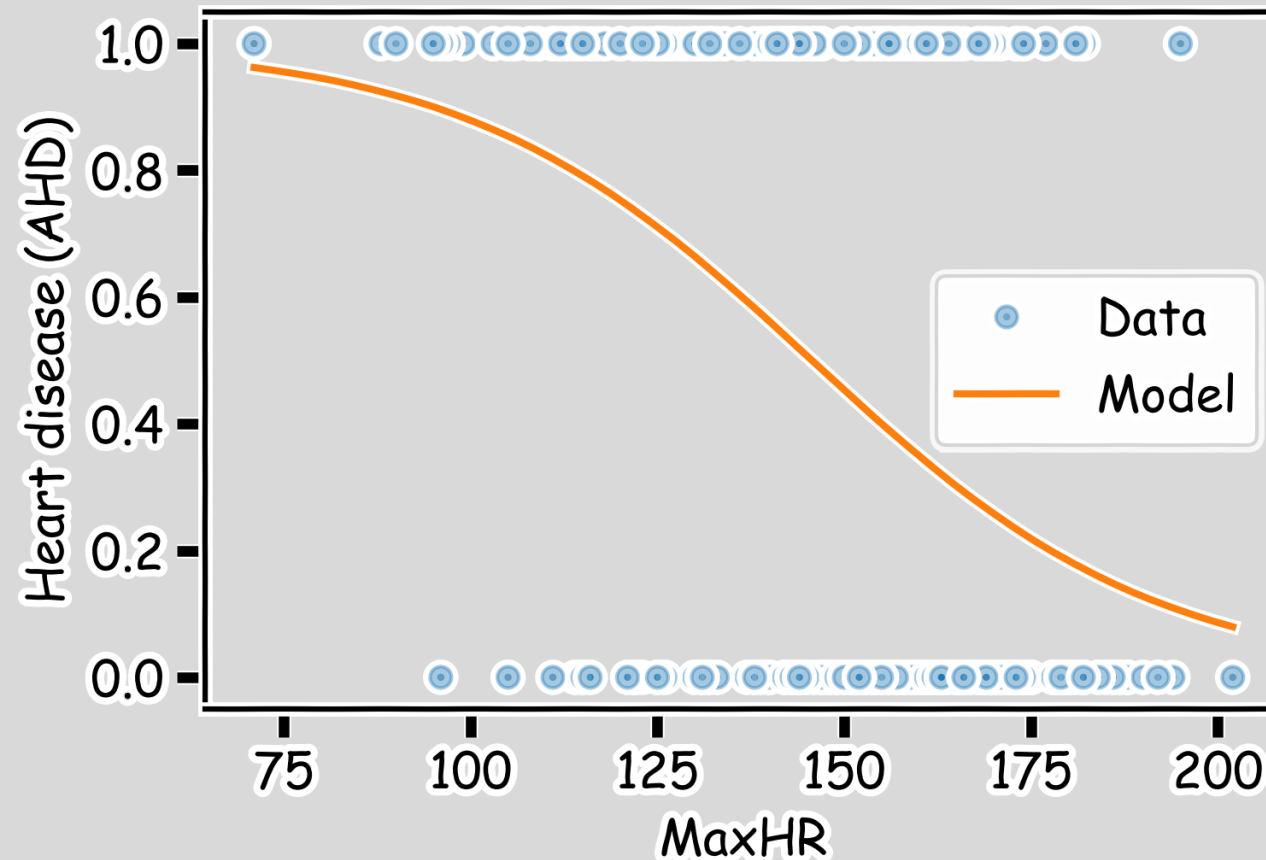
Heart Data: Logistic Estimation

We'd like to predict whether or not a person has a heart disease. And we'd like to make this prediction, for now, just based on the MaxHR.

How should we visualize these data?



Heart Data: Logistic Estimation



Heart Data: Logistic Estimation

There are various ways to fit a logistic model to this data set in Python. The most straightforward in `sklearn` is via `linear_model.LogisticRegression`.

```
from sklearn.linear_model import LogisticRegression

logreg = LogisticRegression(C=100000, fit_intercept=True)
logreg.fit(data_x.values.reshape(-1,1), data_y);

print('Estimated beta1: \n', logreg.coef_)
print('Estimated beta0: \n', logreg.intercept_)
```

```
Estimated beta1:
[[-0.04326016]]
Estimated beta0:
[ 6.30193148]
```



Heart Data: Logistic Estimation

Answer some questions:

- Write down the logistic regression model.
- Interpret $\hat{\beta}_1$.
- Estimate the probability of heart disease for someone with $\text{MaxHR} \approx 200$?
- If we were to use this model purely for classification, how would we do so? See any issues?

Categorical Predictors

Just like in linear regression, when the predictor, X , is binary, the interpretation of the model simplifies (and there is a quick closed form solution).

In this case, what are the interpretations of $\hat{\beta}_0$ and $\hat{\beta}_1$?

For the heart data, let X be the indicator that the individual is a male or female. What is the interpretation of the coefficient estimates in this case?

The observed percentage of HD for women is 26% while it is 55% for men.

Calculate the estimates for $\hat{\beta}_0$ and $\hat{\beta}_1$ if the indicator for HD was predicted from the gender indicator.

Statistical Inference in Logistic Regression

The **uncertainty of the estimates** $\hat{\beta}_0$ and $\hat{\beta}_1$ can be quantified and used to calculate both confidence intervals and hypothesis tests.

The estimate for the standard errors of these estimates, likelihood-based, is based on a quantity called Fisher's Information (beyond the scope of this class), which is related to the curvature of the log-likelihood function.

Due to the nature of the underlying Bernoulli distribution, if you estimate the underlying proportion p_i , you get the variance for free! Because of this, the inferences will be based on the normal approximation (and not t -distribution based).

Of course, you could always **bootstrap** the results to perform these inferences as well.